



Faculty of Computer Science and Mathematics
Universiti Malaysia Terengganu
Email Security and Spam/Phishing Detection
(Verification)

Group11

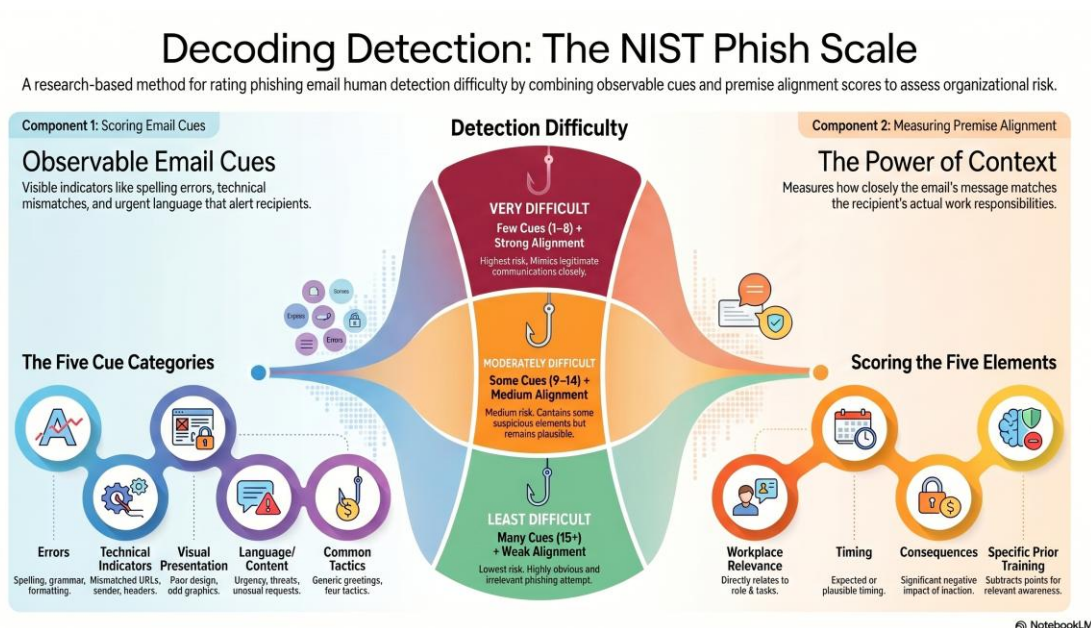
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1. Introduction

The purpose of this verification report is to evaluate the accuracy, completeness, and reliability of the artifacts generated using NotebookLM for the topic *Email Security and Spam/Phishing Detection*. While NotebookLM successfully synthesized information from multiple cybersecurity sources, all generated outputs were manually reviewed and verified against the original references. This process was necessary to ensure that technical concepts related to phishing detection, email security, social engineering, and the NIST Phish Scale were represented accurately and appropriately within the project.

2. Verification of Artifact 1

Figure 1: Decoding Detection – The NIST Phish Scale



Description

This artifact summarizes the NIST Phish Scale framework used to assess the difficulty of detecting phishing emails. The infographic identifies two major components:

- Observable Email Cues
- Premise Alignment

It further categorizes phishing emails according to detection difficulty levels ranging from Least Difficult to Very Difficult.

Verification Against Sources

The content was verified against:

- NIST Technical Note 2276 (Phish Scale User Guide)
- Social Engineering in Phishing and Spear-Phishing Emails
- Microsoft Digital Defense Report 2024

The artifact correctly identifies the five categories of observable cues:

- Errors
- Technical Indicators
- Visual Presentation
- Language and Content
- Common Tactics

These categories are consistent with the NIST Phish Scale framework.

Limitation Identified

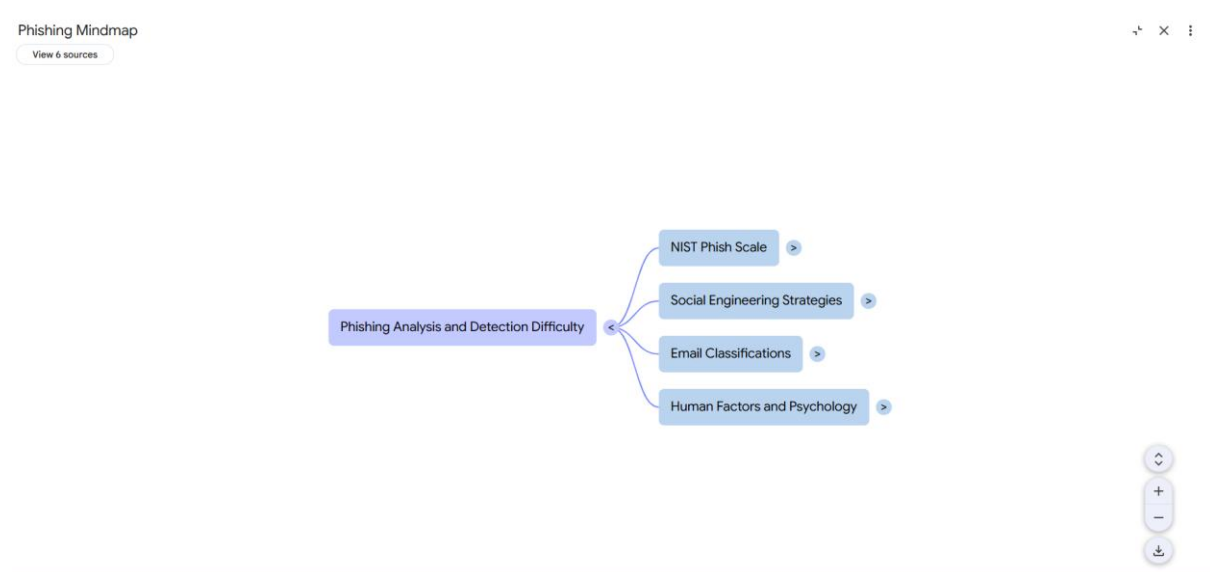
Although the infographic accurately summarizes the NIST model, it simplifies the scoring process. The original NIST methodology involves a detailed calculation process using cue categories and premise alignment measurements. The artifact does not explain how the final score is calculated or how different elements contribute to the overall phishing difficulty rating.

Correction and Improvement

A note should be added stating that the infographic provides a conceptual overview only and that actual NIST scoring requires detailed evaluation criteria described in NIST Technical Note 2276.

3. Verification of Artifact 2

Figure 2: Phishing Mindmap



Description

This artifact presents a mind map of phishing analysis and detection difficulty. The major branches include:

- NIST Phish Scale
- Social Engineering Strategies
- Email Classifications
- Human Factors and Psychology

Verification Against Sources

The categories shown in the mind map are supported by:

- NIST Technical Note 2276
- Social Engineering in Phishing and Spear-Phishing Emails
- Securing Email Entrances and Mitigating Phishing Attacks

The artifact correctly highlights that phishing detection is influenced by both technical and human factors.

Limitation Identified

The mind map does not include several important topics discussed in the uploaded sources, including:

- SPF (Sender Policy Framework)

- DKIM (DomainKeys Identified Mail)
- DMARC
- Machine Learning Detection
- Artificial Intelligence-Based Detection

These topics were heavily discussed in the NIST Trustworthy Email publication and other sources used in the project.

Correction and Improvement

Additional branches should be added to the mind map covering:

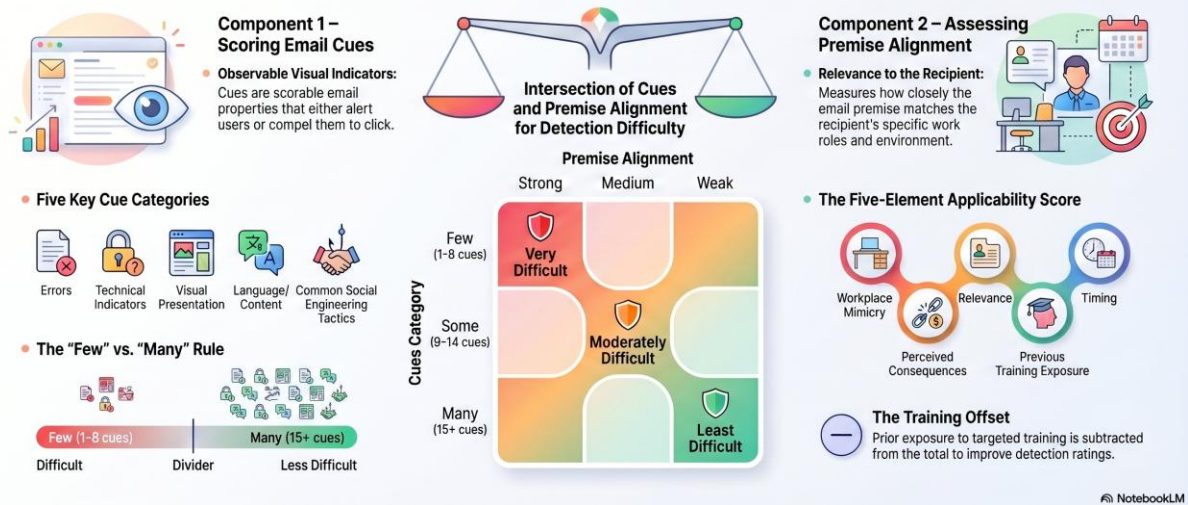
- Email Authentication Protocols
- Machine Learning Approaches
- AI-Based Threat Detection
- Organizational Email Security Measures

This would provide a more comprehensive representation of the topic.

4. Verification of Artifact 3

Figure 3: The NIST Phish Scale – A Practitioner's Guide to Detection Difficulty

The NIST Phish Scale: A Practitioner's Guide to Detection Difficulty



Description

This artifact explains the practical application of the NIST Phish Scale by combining:

- Email cue scoring
- Premise alignment assessment
- Difficulty classification

The infographic illustrates how phishing emails can be categorized according to the number of cues present and the relevance of the phishing scenario to the intended victim.

Verification Against Sources

The information was verified against:

- NIST Technical Note 2276

- Microsoft Digital Defense Report 2024

The classification of:

- Few cues (1–8)
- Some cues (9–14)
- Many cues (15+)

matches the NIST methodology.

Limitation Identified

The artifact may unintentionally suggest that detection difficulty depends solely on observable cues and premise alignment.

However, the uploaded sources emphasize that modern phishing attacks increasingly utilize:

- Artificial Intelligence
- Personalized spear-phishing
- Business Email Compromise (BEC)
- Social engineering techniques

These factors can significantly affect phishing success rates beyond the scope of the NIST scoring model.

Correction and Improvement

An explanatory note should be added indicating that the NIST Phish Scale is a useful assessment framework but should be complemented by modern detection technologies, employee awareness training, and organizational email security controls.

5. Overall Verification Findings

The NotebookLM-generated artifacts accurately summarized the primary concepts found in the uploaded sources and successfully illustrated the NIST Phish Scale framework. However, several simplifications were identified during manual verification.

The main limitations observed include:

- Oversimplification of technical scoring mechanisms.
- Limited discussion of SPF, DKIM, and DMARC.
- Limited representation of machine learning and AI-based phishing detection methods.
- Reduced coverage of current phishing trends such as Business Email Compromise and AI-generated phishing emails.

After manual verification against the six uploaded sources, the artifacts were determined to be generally accurate and suitable for academic use, provided that the identified limitations and improvements are acknowledged.